HW5

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1.

true/false

1. $\{2, \star\} \subseteq \{2, \{\star\}, \{2, \star\}\}$ **False**
2. $\{\{2, \star\}\} \subseteq \{2, \{\star\}, \{2, \star\}\}$ **True**
3. $\{\} \subseteq \{2, \{\star\}, \{2, \star\}\}$ **True**
4. $\star \in \{2, \{\star\}, \{2, \star\}\}$ **False**
5. $\{2\} \subseteq \{2, \{\star\}, \{2, \star\}\}$ **True**
6. $\{\star\} \subseteq \{2, \{\star\}, \{2, \star\}\}$ **False**
7. $\{\star\} \in \{2, \{\star\}, \{2, \star\}\}$ **True**

2.

- A is a set of size 6
- $B = \{1, 2, 3\}$

(a)

$$\mathcal{P}(B) = \big\{\varnothing, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\big\}.$$

(b)

since $|A| = 6$, then the number of subsets of A is $2^{|A|} = 2^6 = 64$ so now we know that $|\mathcal{P}(A)| = 64$

(c)

so we have $|\mathcal{P}(A \cup B)| = 2^{|A \cup B|}$

- the **maximum** happens when A and B are disjoint, so then $|A \cup B| = |A| + |B| = 6 + 3 = 9$ and
$$|\mathcal{P}(A \cup B)|_{\max} = 2^9 = 512$$
- the **minimum** happens happens when $B \subseteq A$, so we have $|A \cup B| = |A| = 6$ and
$$|\mathcal{P}(A \cup B)|_{\min} = 2^6 = 64$$

3.

(a) $A = \{1, 4, 7, 10, 13, \ldots\}$

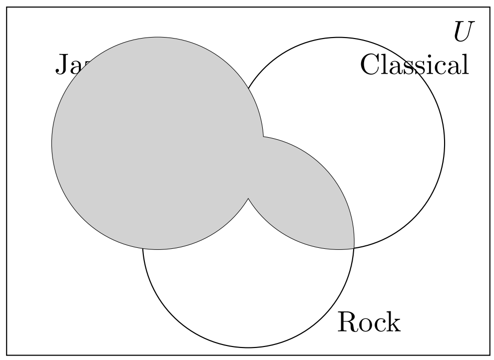
$$A = \{\, x \in \mathbb{Z} \mid \exists k \in \mathbb{Z}\ (k \geq 0 \wedge x = 1 + 3k) \,\}.$$

(b)

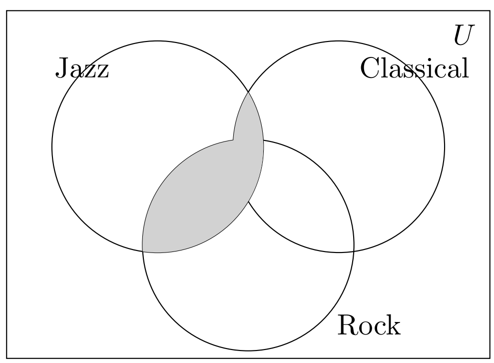
$$\{\, x \in \mathbb{Z} \mid (x < 0) \wedge (x > -300) \wedge \neg \exists q \in \mathbb{Z}\ (x = 5q) \,\}.$$

4.

(a)



(b)



5.

Correct ordering (numbers filled in):

2 $(A - B) \cap B = \{x \mid x \in A \wedge (x \notin B \wedge x \in B)\}$

3 $(A - B) \cap B = \{x \mid x \in A \wedge \text{False}\}$

4 $(A - B) \cap B = \{x \mid \text{False}\}$

0 $(A - B) \cap B = \{x \mid x \in A \vee x \in B\}$

5 $(A - B) \cap B = \{\}$

1 $(A - B) \cap B = \{x \mid (x \in A \wedge x \notin B) \wedge x \in B\}$

6.

$$|A \cup (B \cap C)| = |A| + |B \cap C|.$$

Counterexample: so we have $A = \{1\}$, $B = \{1\}$, and $C = \{1\}$ so then $B \cap C = \{1\}$.

$$|A \cup (B \cap C)| = |\{1\} \cup \{1\}| = |\{1\}| = 1,$$

although..

$$|A| + |B \cap C| = 1 + 1 = 2.$$

so the equality is false in general

7.

$$A \cup B = A \cap B \implies A = B.$$

(i)

lets assume $A \cup B = A \cap B$

- so to show $A \subseteq B$, let $x \in A$, then $x \in A \cup B$ by the assumed equality, $x \in A \cap B$, so $x \in B$.
- right now, to show $B \subseteq A$, we let $x \in B$ then $x \in A \cup B$ by the assumed equality, $x \in A \cap B$, so $x \in A$.

so finally $A \subseteq B$ and $B \subseteq A$, so $A = B$

(ii)

let's assume $A \cup B = A \cap B$
intersect both sides with A^c:

$$(A \cup B) \cap A^c = (A \cap B) \cap A^c.$$

using distributivity on the left we have

$$(A \cup B) \cap A^c = (A \cap A^c) \cup (B \cap A^c) = \varnothing \cup (B \setminus A) = B \setminus A.$$

now on the right

$$(A \cap B) \cap A^c = A \cap A^c \cap B = \varnothing.$$

so now $B \setminus A = \varnothing$, that means $B \subseteq A$
also, intersect both sides with B^c:

$$(A \cup B) \cap B^c = (A \cap B) \cap B^c.$$

so the left side becomes

$$(A \cap B^c) \cup (B \cap B^c) = (A \setminus B) \cup \varnothing = A \setminus B,$$

and the right side is $\varnothing$ so now $A \setminus B = \varnothing$, so $A \subseteq B$.
so now $A \subseteq B$ and $B \subseteq A$, so $A = B$